D 53678 (Pages : 3) Name.....

Reg. No.....

FIRST SEMESTER (CBCSS—UG) DEGREE EXAMINATION NOVEMBER 2023

Mathematics

MTS 1C 01—Mathematics—I

(2019—2023 Admissions)

Time : Two Hours Maximum : 60 Marks

Section A

Answer any number of questions.
Each question carries 2 *marks.*
Maximum Marks 20.

1. Find the tangent line to the parabola $y = x^2 - 3x + 1$ when $x_0 = 2$. Sketch.

2. Find limit if exists, $\lim_{x \to 3} \sqrt{|x-3|}$.

3. Calculate an approximate value for $\sqrt{10}$ using a linear approximation around $x_0 = 9$.

4. Calculate the second derivative of $\frac{1+x}{\sqrt{x}}$.

5. Find the critical points of the function $f(x) = 3x^4 - 8x^3 + 6x^2 - 1$.

6. Find the intervals on which the function $f(x) = \frac{x}{x-1}$ is concave upward and those which they are concave downward,

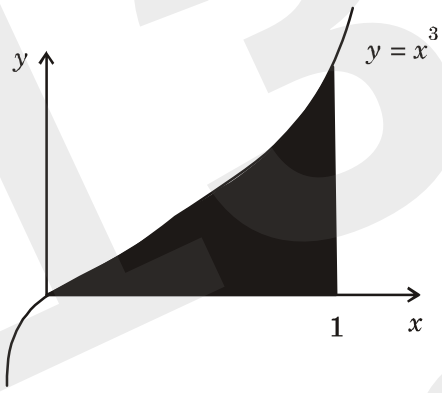
7. A shoe repair shop can produce $2x - x^2 - 3$ dollars of revenue every hour when x workers are employed. Find the marginal productivity when 5 workers are employed.

8. Find $\lim_{x \to 0^+} x \log x$.

9. Find the rate of increase of are of circle with radius r.

Turn over

10. Compute the area of the region shown in Fig.



11. Using the fundamental theorem of calculus, compute $\int_a^b x^2\,dx$.

12. Verify the formula $\frac{d}{dx}\int_a^x f(s)\,ds = f(x)\,f$ or $f(x) = x$.

Section B

Answer any number of questions.
Each question carries 5 *marks.*
Maximum Marks 30.

13. (a) Find $\lim_{x \to 2} -\frac{3x}{x^2 - 4x + 4}$; (b) Find $\lim_{x \to 0} \frac{3x + 2}{x}$.

14. Calculate the linear approximation to the area of a square whose side is 2.01. Draw a geometric figure, obtained from a square of side 2, whose area is exactly that given by the linear approximation.

15. A race car travels mile in 6 seconds, its distance from the start in feet after t seconds being $f(t) = \frac{44}{3}t^2 + 132\,t$. (a) Find its velocity and acceleration as it crosses the finish line ; and (b) How fast was it going halfway down the track ?

16. If $y = f(x)$ and $x^2 + y^2 = 1$ express $\frac{dy}{dx}$ interms of x and y.

17. State Mean Value Theorem .Verify Mean Value Theorem for the function $f(x) = x^3 - 5x^2 - 3x$ in [1, 3].

18. Find the volume of a ball' of radius r.

19. (a) Find the average value of $f(x) = x^2$ on [0, 2].

    (b) Find the volume of the solid obtained by revolving the region under the graph of $3x + 1$ on [0, 2] about the x axis.

Section C

Answer any **one** *question.*
Each question carries 10 *marks.*
Maximum Marks 10.

20. (a) Prove the power rule $\int x^n \, dx = \frac{x^{n+1}}{n+1} + c, n \neq -1.$

    (b) The velocity of a particle moving along a line is $2t + 3$, at a time t. At time 2 the particle is at position 6, where is it at time 15 ?

21. (a) Show that a good approximation to $\frac{1}{1+x}$ when x is small is $1 - x$.

    (b) Find the equation of the tangent line to the curve to $x^6 + y^4 = 9xy$ at the point (1, 2).

(1 × 10 = 10 marks)